Data Preparation Session 3
Exercises

Exercise 3.1.
a) Load the “injuries.csv” file.
b) Explain what is wrong with this data.
c) Make the necessary transformation to have a tidy dataset.
d) Check if the transformation is done, and all known problems are solved. Explain what else to do (it is about thinking and not coding).

Exercise 3.2.
a) Load the “athletes.csv” file. The data is about minimum and maximum heart rates of athletes during 5 tests.
b) Explain what is wrong with this data.
c) Make the necessary transformation to have a better dataset (just one pivot at once).
d) Check if the transformation is done, and all known problems are solved. Explain what should be done as the next step.
e) Try to find how to make the dataset wide. (advanced level, you will learn tricks to do it later)

Exercise 3.3.
a) Import the “sales.csv” dataset. It is about different products’ sales in the months of last year.
b) Make the data tidy using the right pivot function. Hint: (you may have “product”, “month”, and “sales” columns in the final database.

Exercise 3.4.
a) Import the “ratings.tsv” file. It is about average yearly rating of departments in the company during between 2015 and 2024.
b) Prepare your data for charts in Excel: each department should be a column, and each year in a row.

Exercise 3.5.
a) Import the “debt performance.csv” file. Pay attention to the delimiter.
b) Create a version of the data frame in tidy format (each store is a row; each variable is a column).
c) Repeat the exercise with “debt performance2.csv”.

Exercise 3.6.
a) Import the data from the “sales report.xlsx” workbook. Pay attention to the arguments of the function.
b) Make the data tidy. Remember, the database contains sales values for products, in regions, in quarters.
c) Propose additional steps for business analytics (just explain what to do, no coding required here).
d) Create a table with each region and quarter total sales. (advanced level, you will learn tricks to do it later).